

Time Series and Forecast Project: Final Presentation & Delivery Instructions

Purpose

The final presentation is where your group tells the complete story of your project — from the question you set out to answer through to your forecasts and conclusions. By now your analysis should be finished; this session is about communicating it clearly and defending your choices.

What to prepare

Each group will present for 20-25 minutes, followed by 10-20 minutes of Q&A.

1. Question and motivation (brief)

Restate the question you forecast or investigated and why it matters. Keep this short — one or two slides. Assume the audience remembers the broad topic from your progress update.

2. Data

Summarize the dataset: source, frequency, sample period, and number of observations. Show the main time plot and point out the features that shaped your modeling (trend, seasonality, breaks, outliers). If you changed or extended your data since the milestone, say so.

3. Methodology

Walk through what you did and *why*, in a logical sequence:

- Transformations and their justification (log, differencing, seasonal adjustment).
- Stationarity testing (ADF, KPSS) and how it informed your modeling.
- Model identification — how you used ACF/PACF, information criteria (AIC/BIC), or other reasoning to arrive at your specification.
- The final model(s) you estimated. If you compared several, explain the comparison.

4. Results and diagnostics

This is the core of the talk. Present:

- Your estimated model with key coefficients and what they mean substantively

- Residual diagnostics (Ljung-Box, residual ACF, normality) and an honest read of whether assumptions hold.
- Forecasts with prediction intervals, plotted against actuals where possible.
- Forecast evaluation — out-of-sample accuracy using RMSE, MAE, MAPE, or a comparison against a benchmark (e.g., a naïve or seasonal-naïve forecast). A model that beats no benchmark is a result worth stating plainly.

5. Conclusions and limitations

State what you found, what it means, and where your approach falls short. Acknowledging limitations — short sample, structural change, assumptions that didn't fully hold — earns more credit than overclaiming. End with one or two sentences on what you'd do with more time or data.

Format and logistics

- Slides in any software. Aim for roughly 20–30 slides. Put detailed regression output, full diagnostic panels, and code in a backup appendix rather than the main flow.
 - Every group member must present a meaningful portion. Grades include an individual component, so plan speaking roles deliberately.
 - Be ready to pull up your code, notebook, or full output during Q&A. You won't present code on slides, but you should be able to navigate to any result if asked.
 - Practice for timing. Going significantly over will cost you — a tight, well-rehearsed talk reads as command of the material.
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Deliverables and deadlines

Submit the following to the course portal by [start of class on presentation day / stated deadline]:

1. **Slide file** (PDF or ppt).
2. **Reproducible code** — your script(s) or notebook, runnable from the submitted data (or snippet if the data is too big), with a short README noting any package requirements.
3. **Data file or a documented link** to the source.
4. **Written report** (PDF).

Name files clearly: `Group##_TimeSeries_[component]`. One submission per group is sufficient.

Evaluation criteria

The final presentation and deliverables are graded on five dimensions:

1. **Technical correctness** — Are the methods applied and interpreted correctly?
2. **Analytical depth** — Did the group go beyond mechanical steps to reason about their data and results?
3. **Communication** — Is the talk clear, well-paced, and well-structured? Are visuals legible and purposeful?
4. **Reproducibility** — Can the instructor rerun the analysis from the submitted materials?
5. **Q&A and individual contribution** — Can each member explain and defend the work?

The final presentation and deliverables count for **50%** of the total grade.

Common pitfalls to avoid

- *Dumping output.* Every table and plot needs interpretation. If a slide has a number on it, you should be able to say in one sentence why it matters.
- *Skipping forecast evaluation.* A forecast with no accuracy assessment and no benchmark is incomplete. Always show how good (or not) your model actually is.
- *Hiding problems.* If your residuals fail a test or your model underperforms a naïve benchmark, present it honestly and discuss why. Graders reward diagnosis over concealment.
- *Uneven delivery.* A presentation where one member clearly did the analysis and the others read slides will hurt the group. Make sure everyone understands the whole project.
- *Non-reproducible code.* Test that your submitted code runs cleanly on a fresh machine before submitting.